RAINCAST

PRESENTED BY ADRIAN BUBUIANU



Origin of the creative idea

- Estimating rainfall → fundamental **challenge** in global hydrology and atmospheric science

- Satellites cannot **directly** measure precipitation → rainfall must be **inferred** from **radiative signals**;

- This indirect approach introduces **uncertainties** especially in zones with special climates such as **coastal areas**



Multi-Sensor Synergy

I combined IR, GPM-GMI microwave, SMAP land surface, and reanalysis datasets (MERRA-2, ERA5)

Consistency

I used data spanning 2018–2024 and incorporated historical information through feature windows.

My approach

Project vision and mission



Multiple raw sources, hundreds of GBs of data ⇒ Final aggregated dataset: ~40 MB (CSV format)



Satellite data: For maximum information ⇒ identified the nearest pixel to the New Orleans station



Irregular timestamps ⇒ Enabled correlation across satellite datasets from different sources

Ground Truth









Global Historical Climatology Network-Hourly

- **Station**: NEW ORLEANS LAKEFRONT AP
- **Resolution**: Hourly data
- **Measurement**: Precipitation amount
- **Format**: CSV



Challenge

Missing data

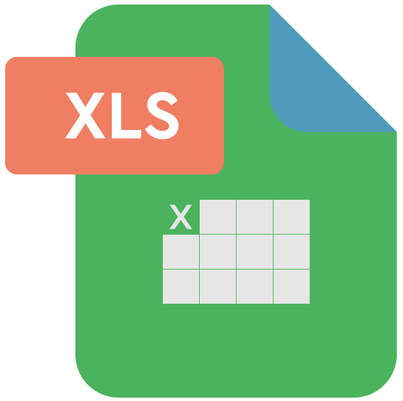
- Required extensive data cleaning, resulting in a reduced dataset.



– GPM-GMI 2024 data –

Update App

S1_frozenPr	S1_cloudW	S1_totalCol	S1_probabi	S1_mostLik	S1_surface	S1_rainWat	S1_convect	S1_airmass	Time
0	0.016496	10	3	0.002015	14	0.056796	0.001471		1/1/2024 5:00
0	0.027795	22	4	0.034265	14	0.023592	0.000994		1/1/2024 15:00
0.000114	0.0151	7	3	0.002764	14	0.054401	0.002051		1/2/2024 5:00
-9999	-9999	9		-9999	4	-9999	-9999		1/2/2024 6:00
0.000111	0.009758	5	3	0	14	0.041221	0.001199		1/2/2024 15:00
-9999	-9999	11		-9999	1	-9999	-9999		1/3/2024 4:00
-9999	-9999	15		-9999	4	-9999	-9999		1/3/2024 6:00
-9999	-9999	13		-9999	3	-9999	-9999		1/3/2024 14:00
-9999	-9999	18		-9999	1	-9999	-9999		1/3/2024 16:00
-9999	-9999	27		-9999	15	-9999	-9999		1/4/2024 3:00



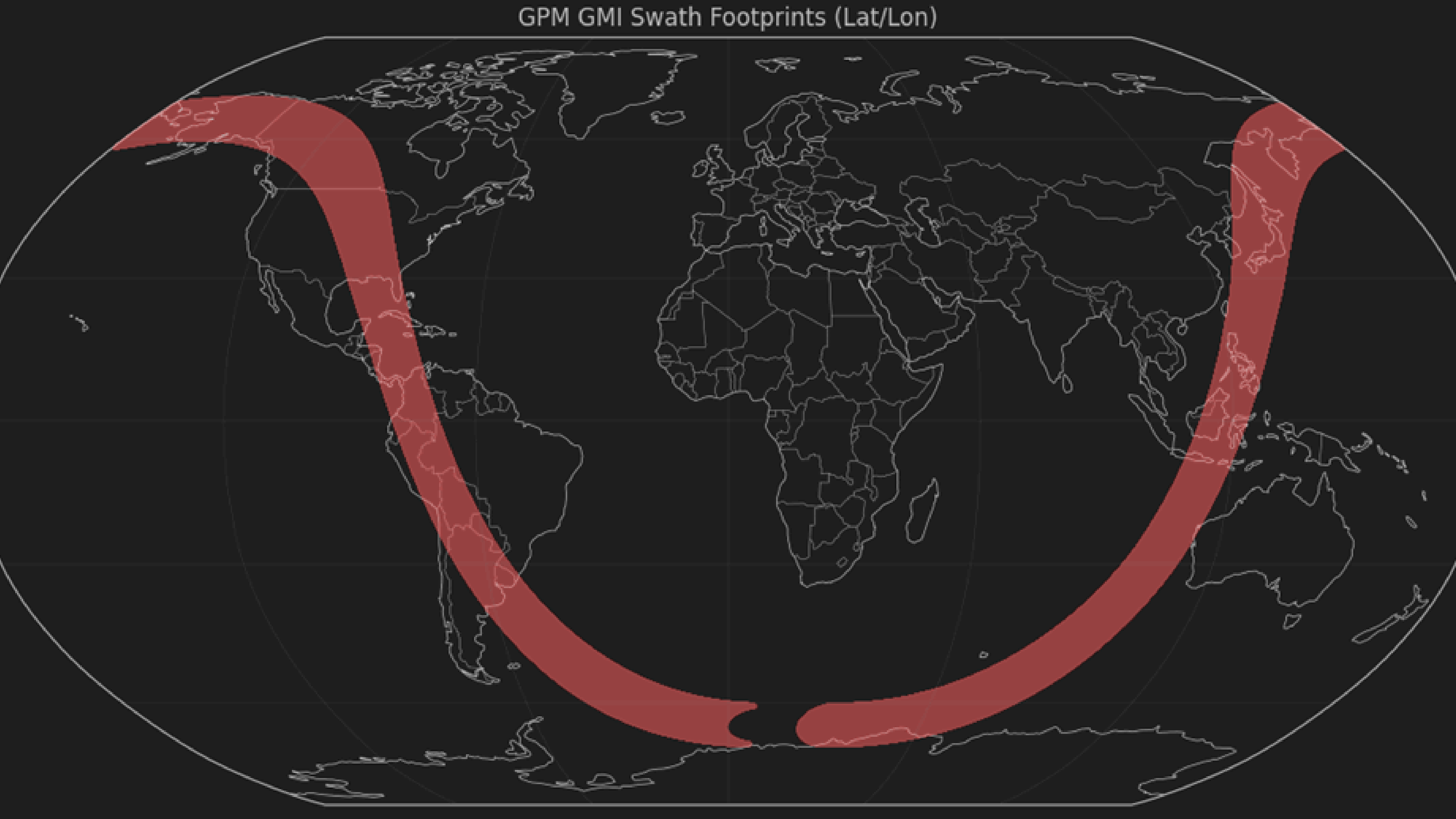
Data formats & Processing

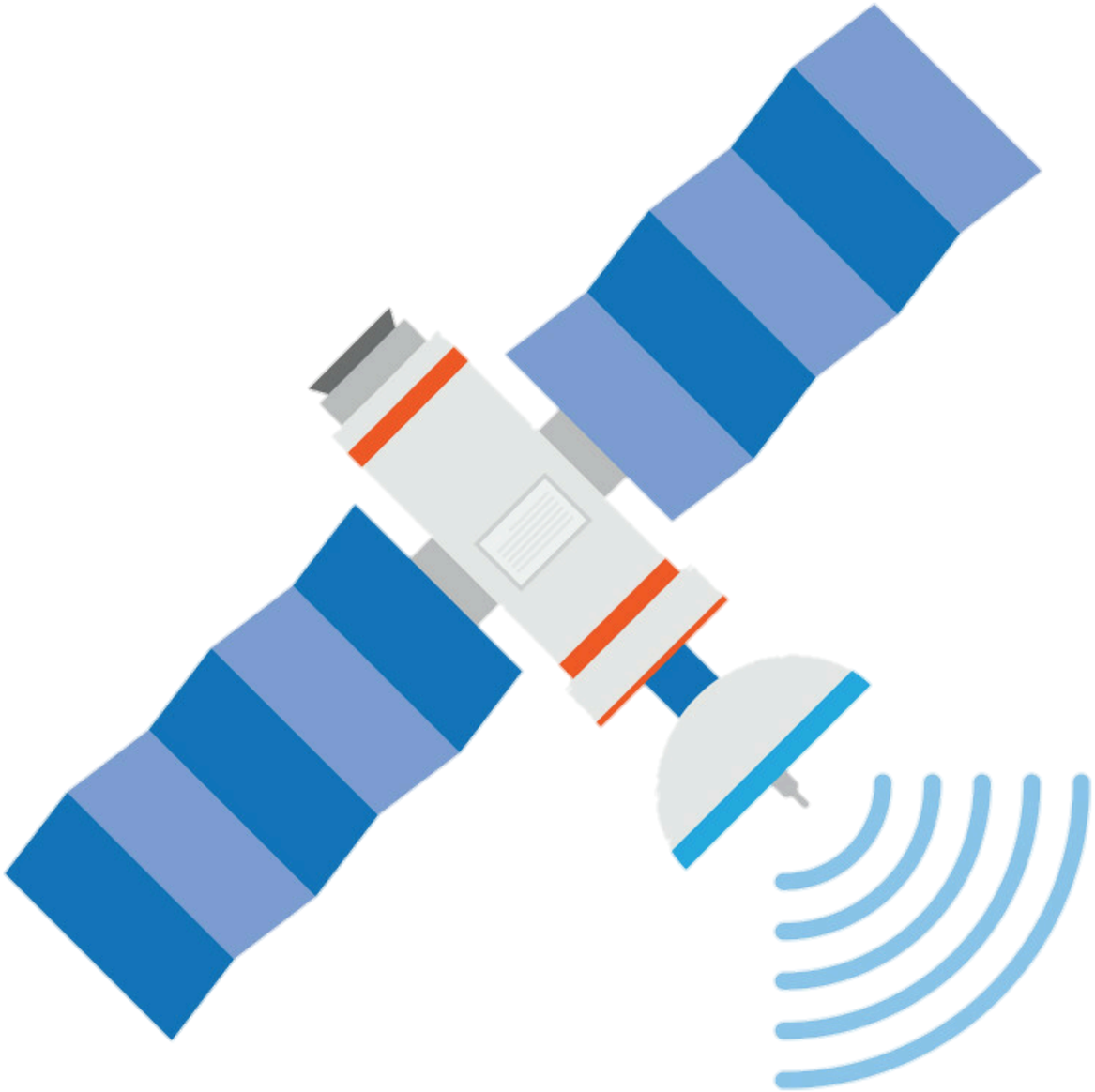


netCDF format	Python library: netCDF4 → Identify variables and attributes, correlate with latitude/longitude to extract values
HDF format	Challenge (requires Miniconda) Python library: pyhdf.SD
HDF5 format	Python library: h5py → Identify variable groups and variables within each group, correlate to lat/lon for values
GRIB format	Python library: xarray



Datasets

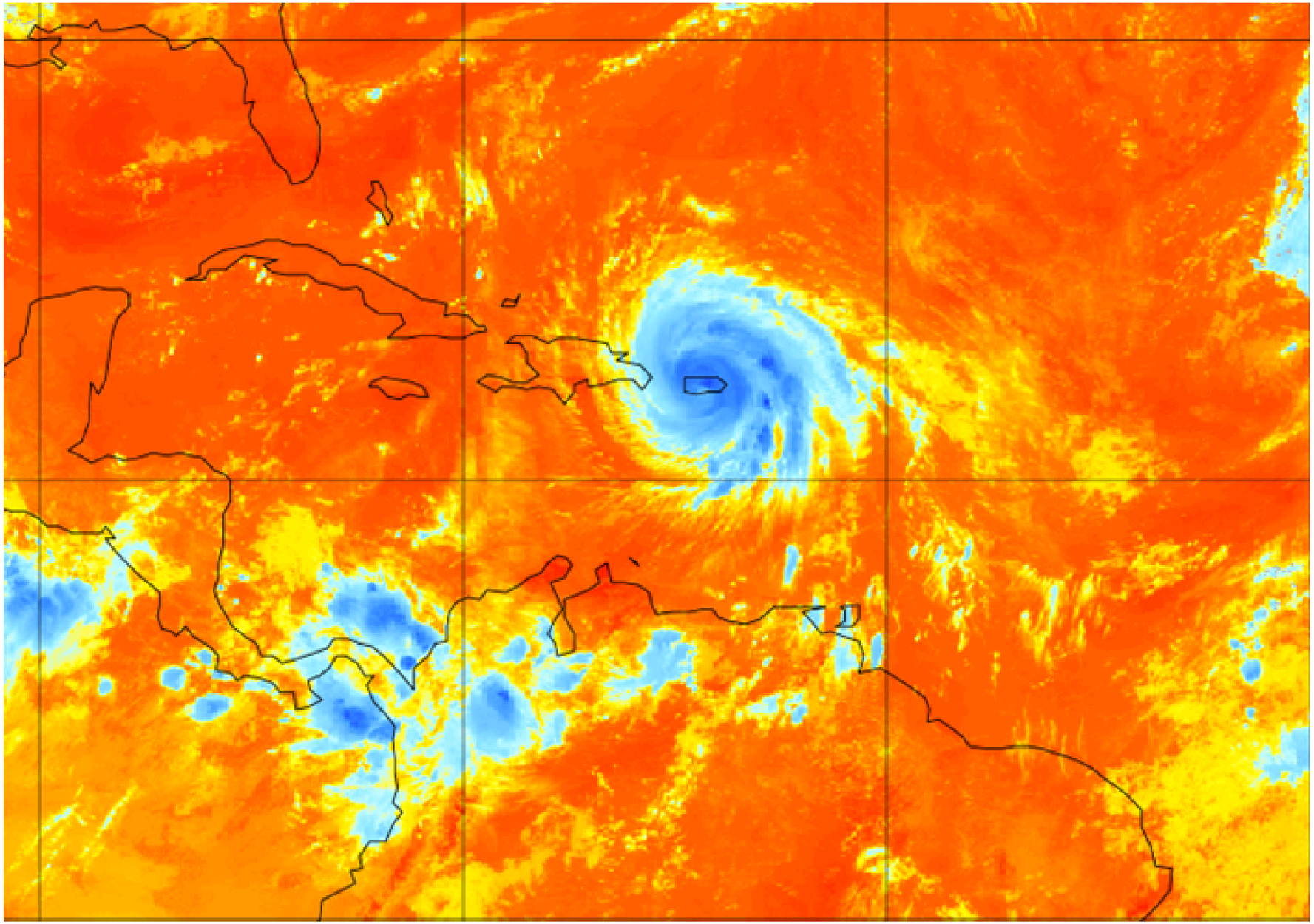

GPM GMI Swath Footprints (Lat/Lon)



GPM GMI (GPROF)

Radiometer Precipitation Profiling

- **Satellites**: GPM Core Observatory (GMI) plus the GPM constellation
- **Temporal Coverage**: Swaths every 1.5 hours, but typically ~daily data for New Orleans
- **Format**: HDF5
- **Spatial Coverage**: Not global! Data only available along GPM satellite tracks
- **Key Features**: surfacePrecipitation, frozenPrecipitation, cloudWaterPath, totalColumnWaterVaporIndex, probabilityOfPrecip, mostLikelyPrecipitation, surfaceTypeIndex, rainWaterPath, convectivePrecipitation, airmassLiftIndex



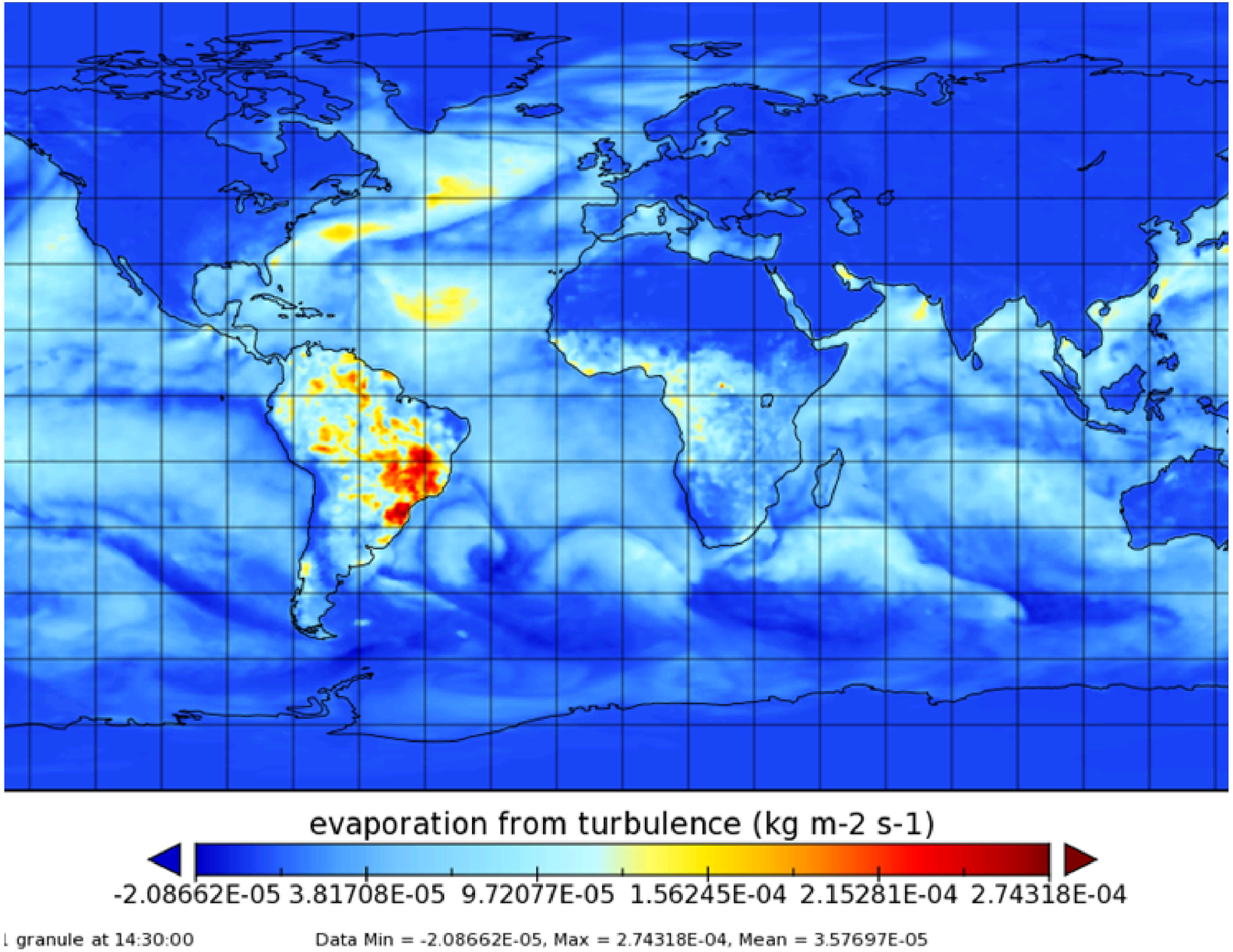
IR GPM MERGIR NCEP

Source: NOAA/NCEP

Format: netCDF

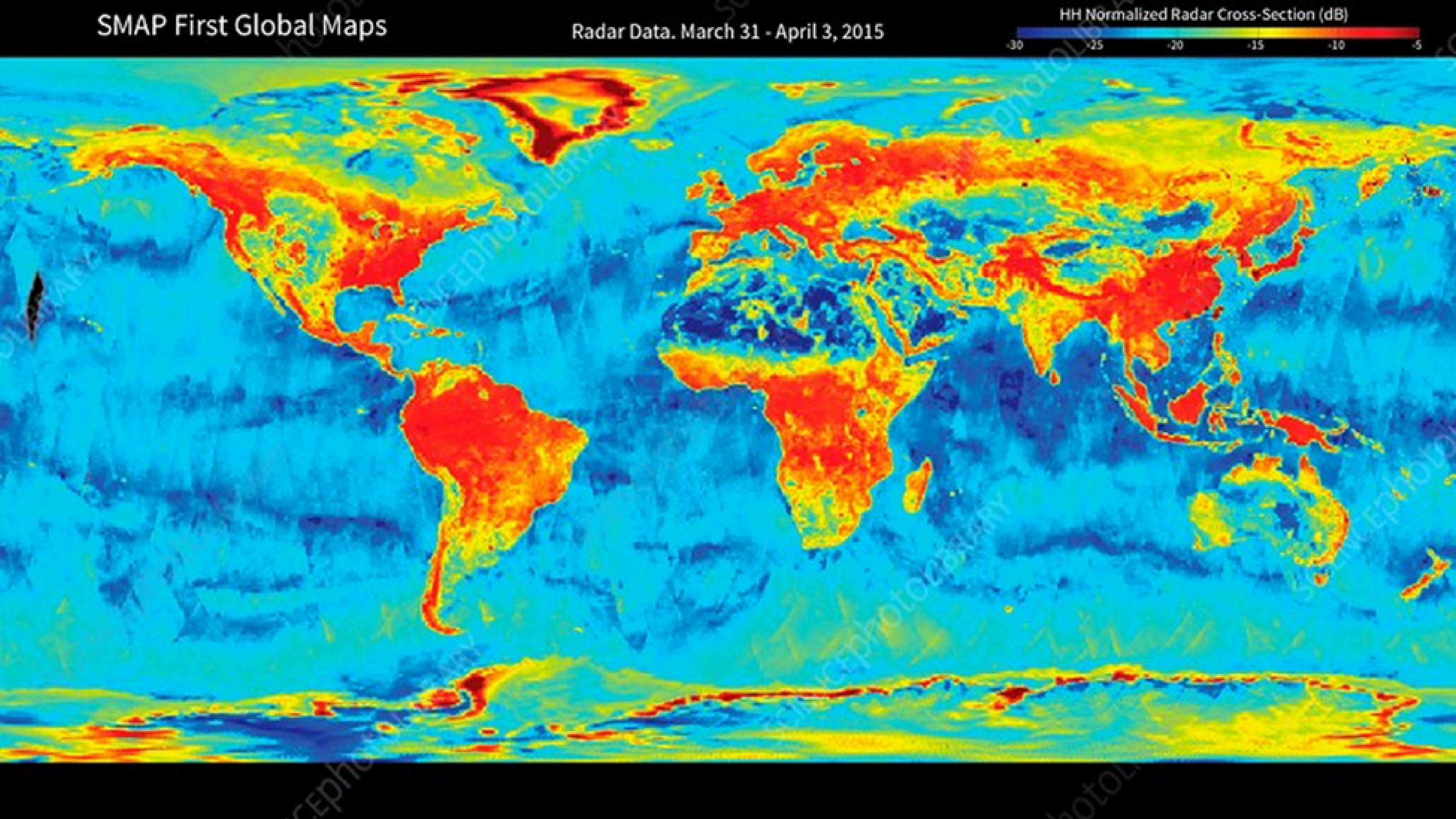
Temporal coverage: Half-hourly

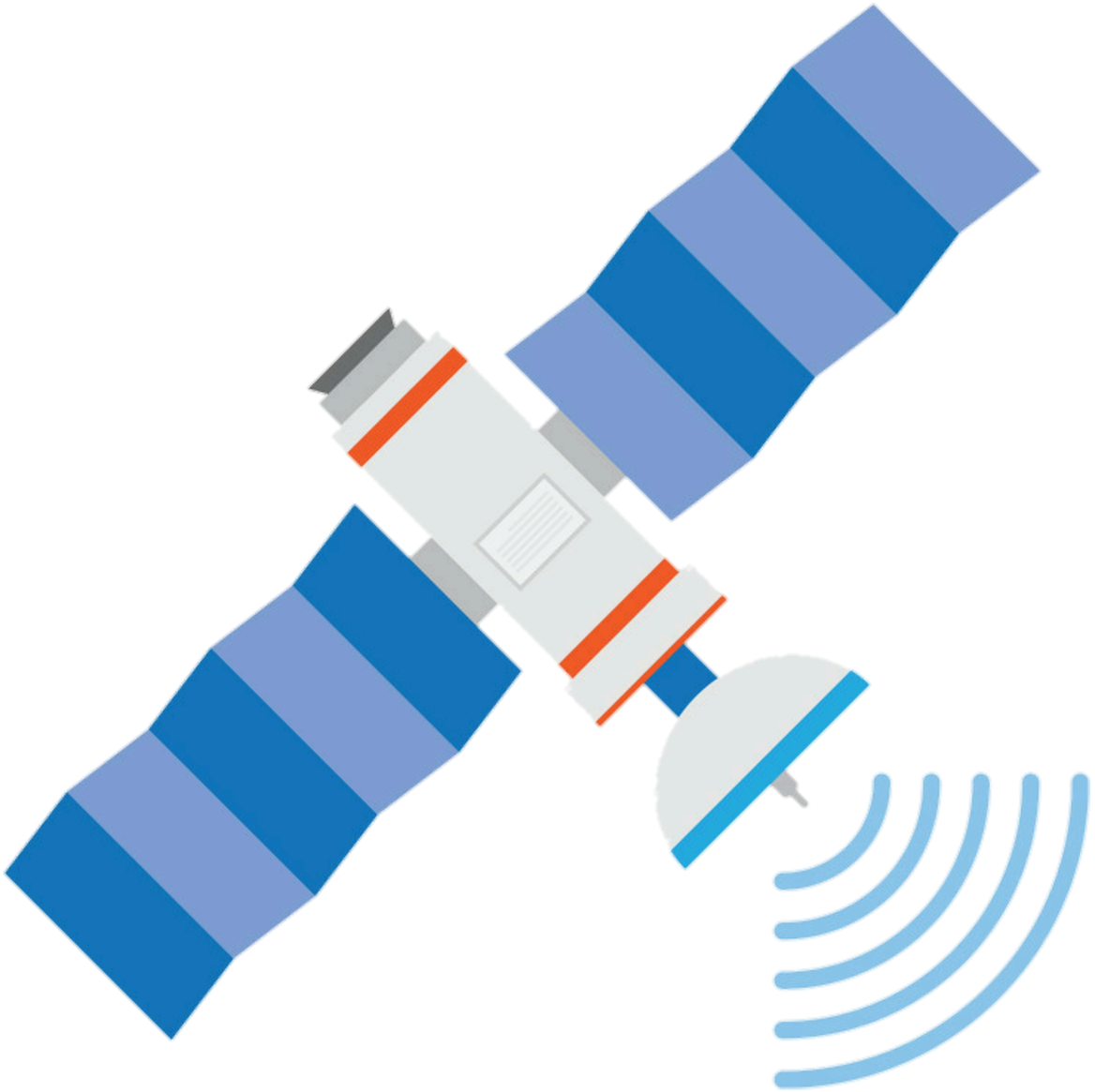
Key Feature: Brightness temperature (Tb) values



MERRA-2 Reanalysis Data

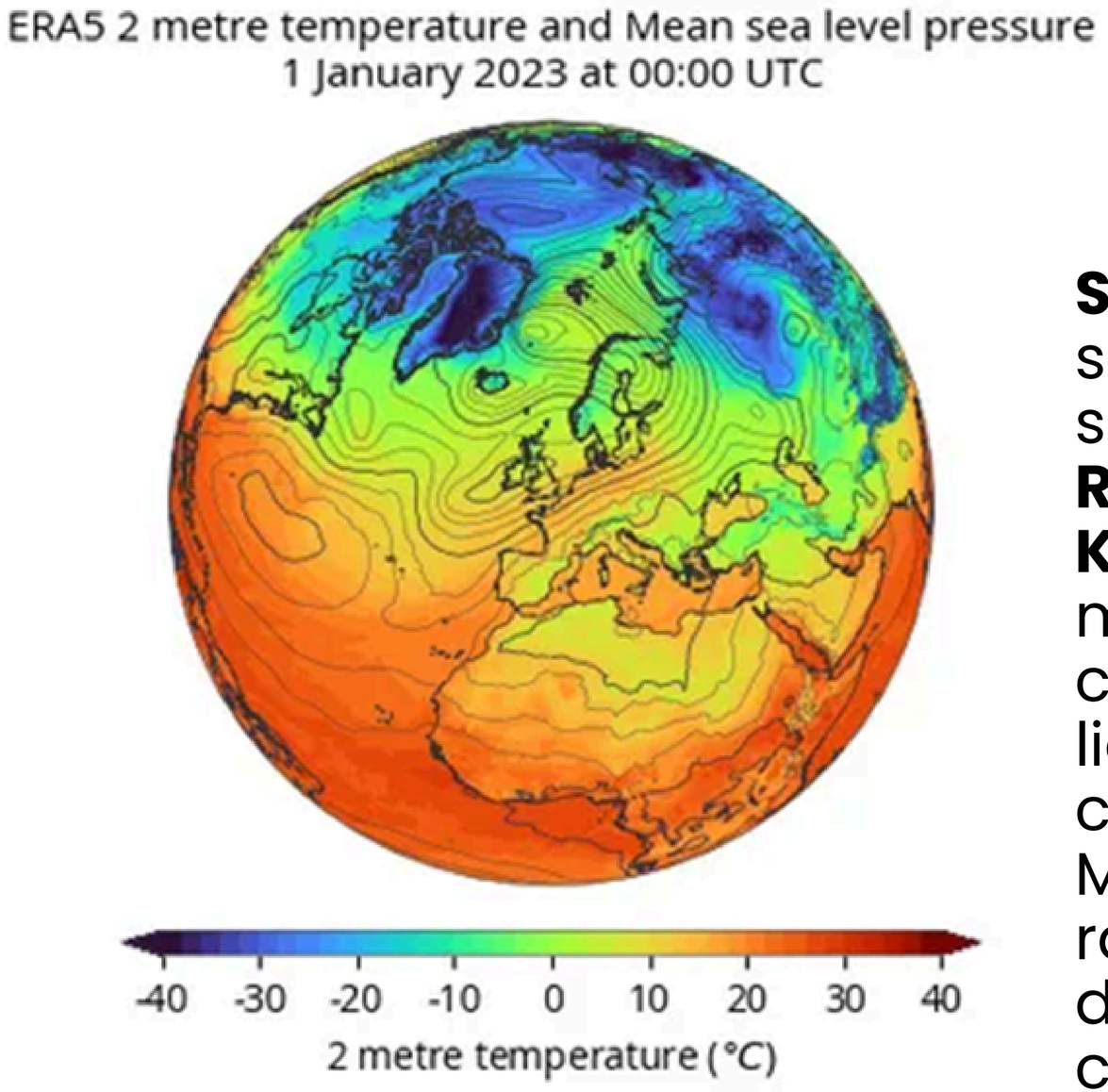
Resolution: Hourly
Source: Not a single satellite; reanalysis uses multiple satellite sources
Format: netCDF
Key features: EFLUX, EVAP, FRCAN, FRCCN, FRCLS, HFLUX, HLML, PBLH
PRECANV, PRECCON, PRECLSC, PRECTOT, QLML, RISFC
TLML, ULML, USTAR, VLML


SMAP First Global Maps
Radar Data. March 31 - April 3, 2015
HH Normalized Radar Cross-Section (dB)
-30
-25
-20
-15
-10
-5



SMAP (Soil Moisture Active Passive)

Satellite: SMAP
Format: HDF5 (.h5)
Resolution: Daily
Key features: soil_moisture, surface_temperature, static_water_body_fraction, radar_water_body_fraction, vegetation_opacity, vegetation_water_content, albedo, tb_h_corrected, tb_v_corrected



ERA5 (Copernicus)

Source: Reanalysis (not a single satellite); uses data from multiple satellites
Resolution: Hourly
Key features: Vertically integrated moisture divergence, u10, v10, Total column rain water, Total column cloud liquid water, CAPE, Total cloud cover, Low cloud cover, Mean vertically integrated, Moisture divergence, Mean evaporation rate, Mean sea level pressure, 2m dewpoint temperature, TCWV (Total column water vapor)



AI Models & Approaches

Hurdle model

Two-stage ablation - Tweedie regressor

Hurdle Model for Current Hour Precipitation

- **Target**: Predicts current hourly rainfall amount using input features and lagged precipitation values (at 1, 3, 6, and 24 hours)



Selected Features
TCWV
vimd
cloudWaterPath
tcc
Tb
Temperature
surface temperature
CAPE
PBLH
RISFC
v10
soil moisture
EFLUX

Metrics (mm/h)

Validation all

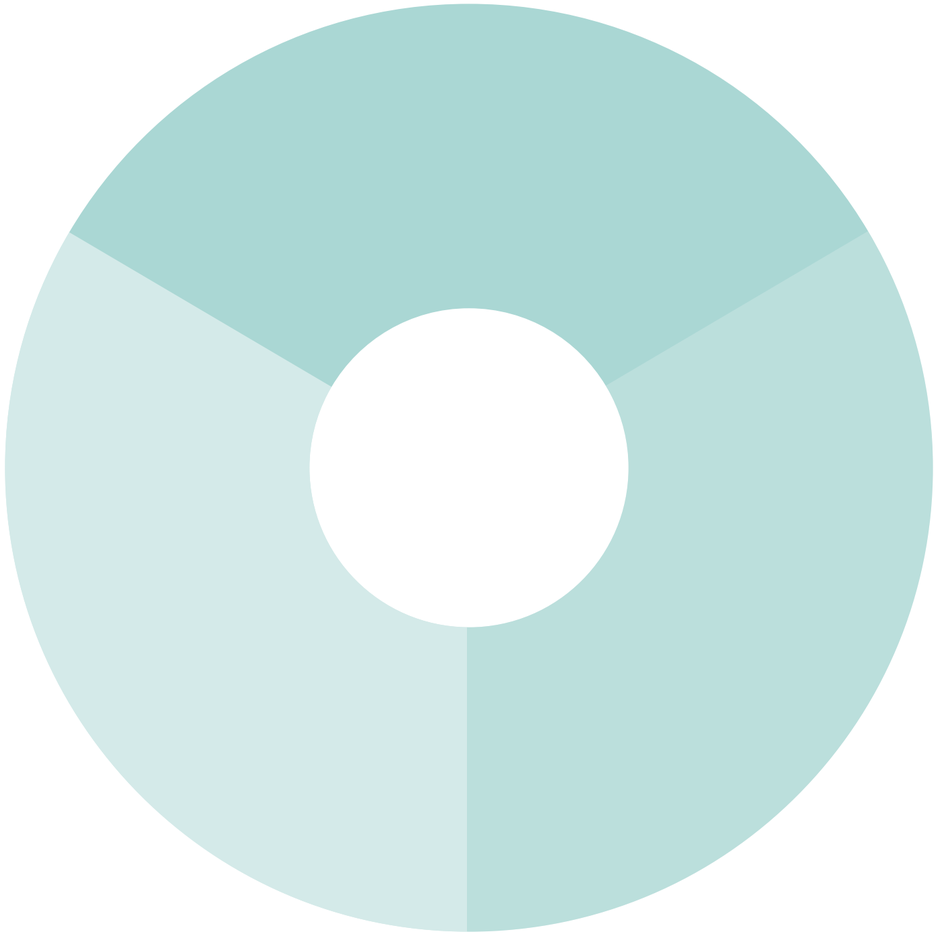
0.1038

MAE

Validation positive

2.9140

MAE



Test all

0.2055

MAE

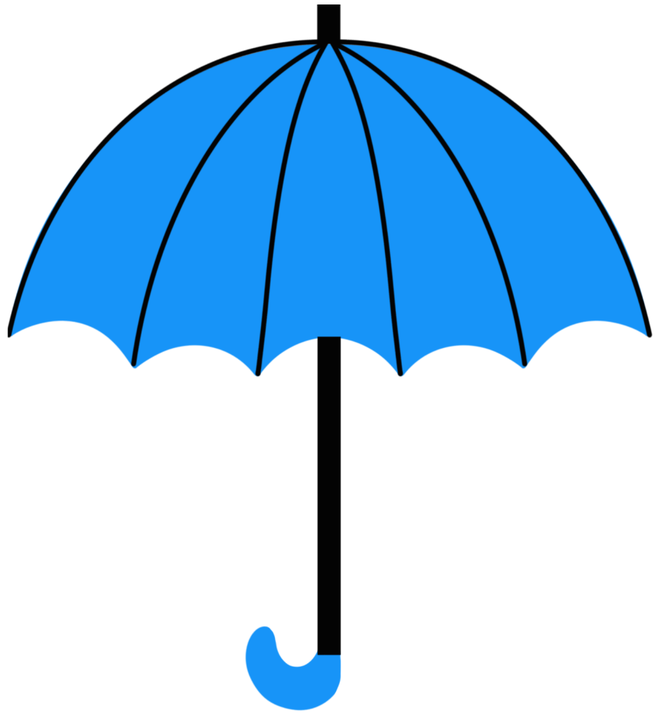
Test positive

2.7645

MAE

Test Predictions vs Ground Truths

Timestamp	Ground Truth	Predicted Amount
2024-01-01 05:00:00	0	0
2024-01-01 06:00:00	0	0
2024-01-01 07:00:00	0	0
2024-01-01 08:00:00	0	0
2024-01-01 09:00:00	0	0
2024-01-01 10:00:00	0	0.002382
2024-01-01 11:00:00	0	0.000927
2024-01-01 12:00:00	0	0.002894
2024-01-01 13:00:00	0	0.064941



Test Predictions vs Ground Truths Rainy Hours Only

Timestamp	Ground Truth	Predicted Amount
2024-01-03 05:00:00	0.8	0.000159
2024-01-03 12:00:00	1	0.925838
2024-01-03 13:00:00	2.8	1.248797
2024-01-03 14:00:00	3.3	3.194189
2024-01-03 15:00:00	2.5	2.596674
2024-01-05 21:00:00	0.3	1.458551
2024-01-05 22:00:00	5.3	2.379185
2024-01-05 23:00:00	2.5	3.024718
2024-01-06 00:00:00	0.3	1.878672

Next Hour Precipitation Prediction Model

Best performing model: Two-stage ablation (Classifier + Tweedie Regressor)



Category	Features Used
Target & Historical (Lags)	Precipitation
Humidity & Cloudiness	TCWV, d2m, tcclw, tcrw, tcc, lcc, ir_Tb, S1_cloudWaterPath, S1_iceWaterPath, S1_totalColumnWaterVaporIndex
Thermodynamics & Instability	CAPE, vimd, msl, Temperature, surface_temperature, m_EFLUX, m_HFLUX, m_PBLH
Wind	u10, v10, m_USTAR
Time (Cyclic)	hour_sin, hour_cos, doy_sin, doy_cos
Extra	soil_moisture, albedo, vegetation_water_content

Modeling Workflow Summary

Features Used for Lags (window: 12h)
Precipitation
TCWV
msl
Tb
v10, u10
CAPE
Cloud Water Path



→ Two-step approach:

1. **Classification** (LightGBM)
   - Hyperparameters optimized via random search
2. **Regression** (Positive-only Tweedie Regressor)
   - Hyperparameters optimized with Optuna

Note: Classification threshold was selected dynamically

Metrics (mm/h)

Validation all

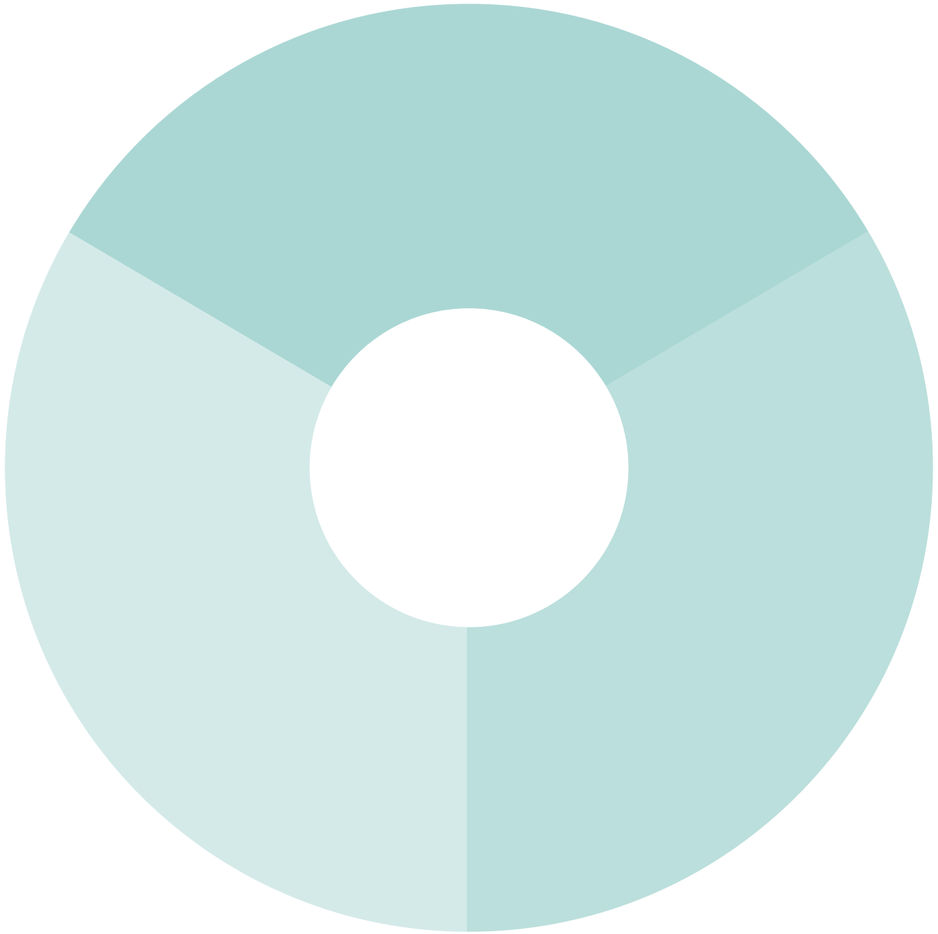
0.155

MAE

Validation positive

3.251

MAE



Test all

0.277

MAE

Test positive

2.918

MAE

Sample predictions

Time	Ground Truth	Probability	Next Hour Precipitation
2024-12-31 13:00:00	0	0.000402	0
2024-12-31 14:00:00	0	0.00046	0
2024-12-31 15:00:00	0	0.000225	0
2024-12-31 16:00:00	0	0.000194	0
2024-12-31 17:00:00	0	0.000086	0
2024-12-31 18:00:00	0	0.000038	0
2024-12-31 19:00:00	0	0.000036	0
2024-12-31 20:00:00	0	0.000027	0
2024-12-31 21:00:00	0	0.000059	0





Sample predictions on rain events

Time	Ground Truth	Next Hour Precipitation
2024-12-27 10:00:00	0.8	0.798573
2024-12-27 11:00:00	0.8	1.597959
2024-12-27 13:00:00	7.9	1.285234
2024-12-28 10:00:00	0.3	0.531142
2024-12-28 13:00:00	1.8	0.92052
2024-12-28 15:00:00	2.3	0.670917
2024-12-28 19:00:00	0.8	0.463792
2024-12-29 03:00:00	14.7	5.936087
2024-12-29 04:00:00	4.6	4.937827
2024-12-29 05:00:00	0.5	1.483616

My Model vs. GPM-GMI Comparison

Time	GPM-GMI	My model	Actual
10/3/2024 22:00	2.191332	0.951088	1
9/26/2024 0:00	0.746303	0	0
9/12/2024 3:00	0.282228	4.162504	2.3
9/4/2024 16:00	0.206942	2.859129	17.5
7/21/2024 19:00	3.406391	0.815159	0.3
7/20/2024 19:00	0.975853	1.739463	0.3
6/28/2024 2:00	2.951165	0	0
6/2/2024 19:00	2.010499	2.254291	0.5
5/14/2024 1:00	1.730762	2.282202	19.6
1/9/2024 13:00	0.188746	0	0

S.W.O.T

swot analysis

S

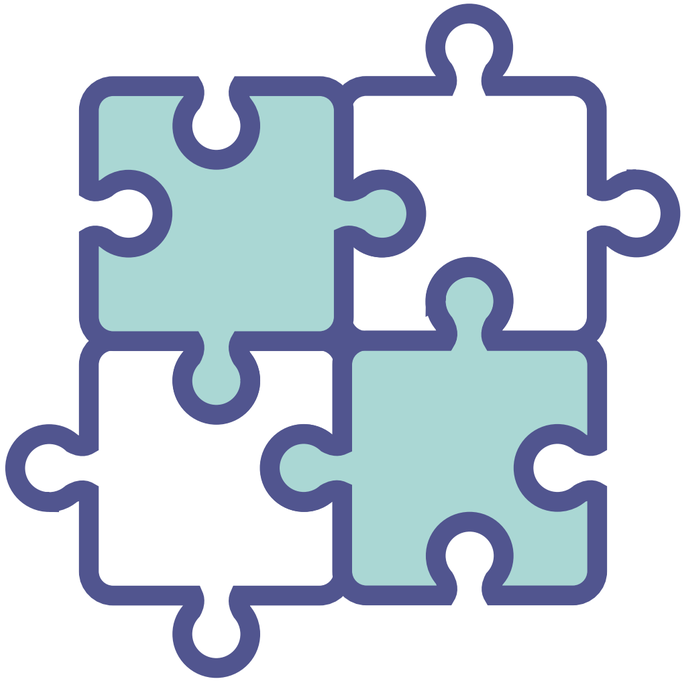
Strengths

- **Early rainfall prediction**: Provides opportunities for public warnings and damage prevention.

W

Weaknesses

- **Insufficient diversity**: Trained solely on New Orleans data (6 years), the model may struggle to generalize across different weather patterns.



T

Threats

- **Risk of inaccurate predictions**
- **Potential impact**: Incorrect forecasts could lead to damages or operational issues.

O

Opportunities

- Extend to **~20 years** of data
- Potential to incorporate **multiple weather stations**
- Option to add **other climatic factors** such as CLDPROP or AIRS

Thank you very much!

PRESENTED BY ADRIAN BUBUIANU

